

YeshwantraoChavan College of Engineering

(An Autonomous Institution affiliated to RashtrasantTukadojiMaharaj Nagpur University)

Accredited 'A++' Grade by NAAC

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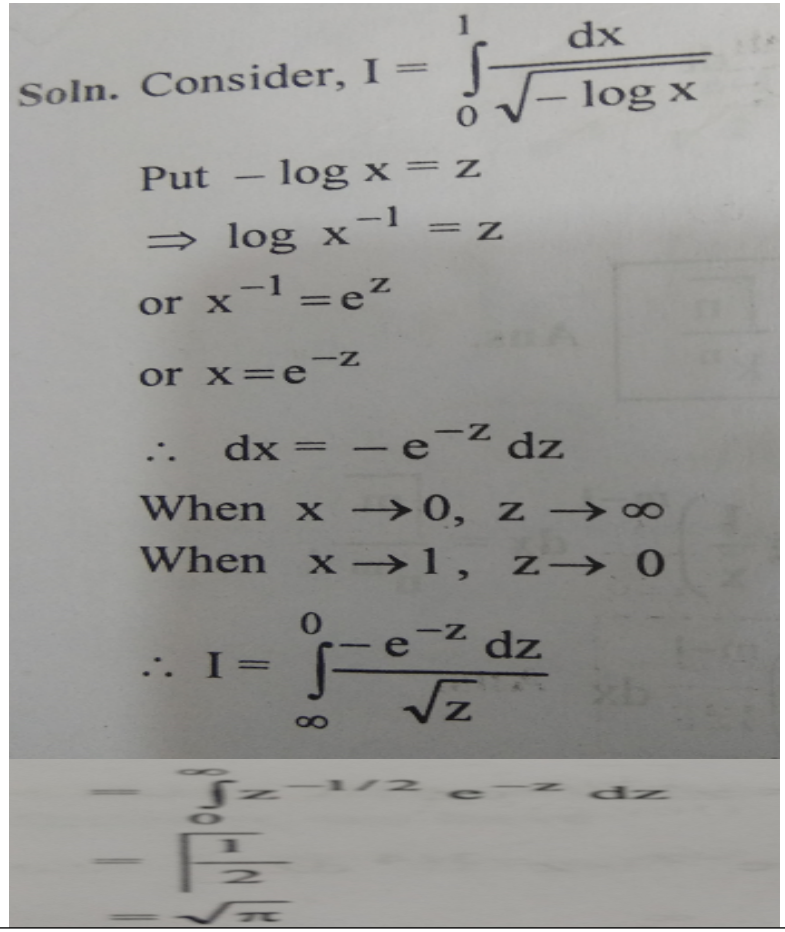
Department of Applied Mathematics & Humanities

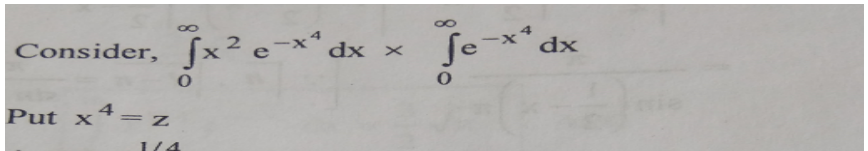
Electronic Question Answer Bank

Course Name: Calculus and Vector (C & V)

Course Code: 23GE1101

Unit-III: Integral Calculus

Sr.No.	Text of Question	Time required in Minutes	Marks allotted	lengthiness	Difficulty level	CO Mapping	Type Descriptive / Numerical
1	Evaluate $\int_0^1 \frac{dx}{\sqrt{-\log x}}$ 	15	5	Medium	Easy	CO3	Descriptive
2		15	5	Medium	Easy	CO3	Descriptive

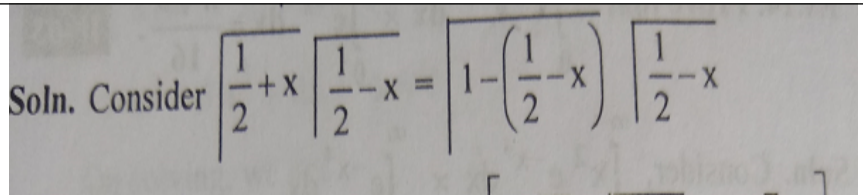


	Prove that :						
		15	5	Medium	Easy	CO3	Descriptive

Soln. Consider, $I = \int_0^\infty x^{1/4} e^{-\sqrt{x}} dx$

Put $\sqrt{x} = z$ or $x = z^2$

3	Evaluate dx						
4	Prove that $\Gamma\left(\frac{1}{2}+x\right)\Gamma\left(\frac{1}{2}-x\right)=\frac{\pi}{\cos \pi x}$	15	5	Medium	Easy	CO3	Descriptive

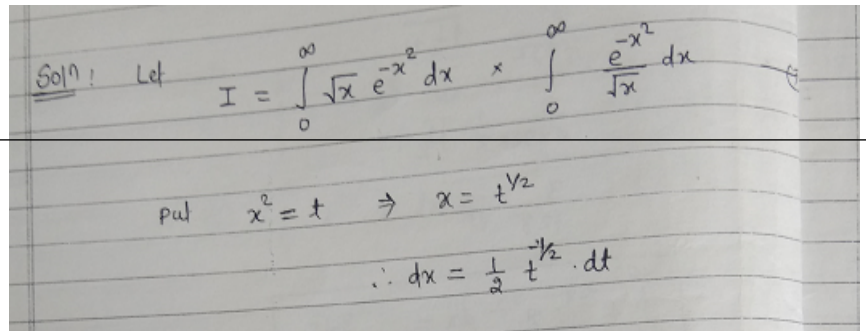


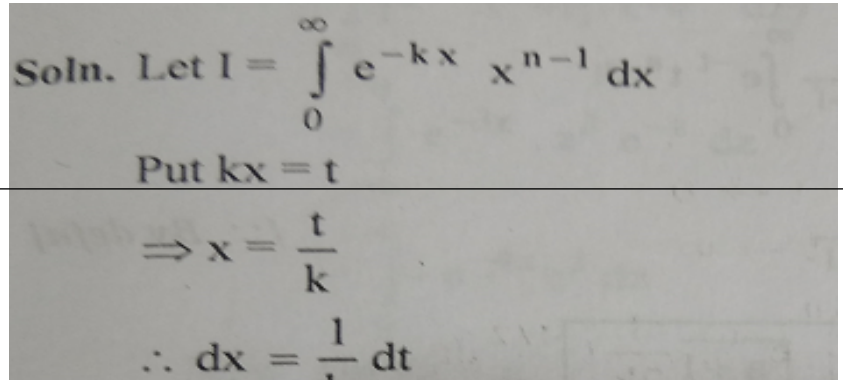
5	Evaluate : $\int_0^{\infty} e^{-x^2} x^2 dx$	15	5	Medium	Easy	CO3	Descriptive

Soln. Consider, $\int_0^{\infty} e^{-x^2} x^2 \, dx$

Put $x^2 = z$

$\Rightarrow x = z^{1/2}$

6	Prove that $\int_0^\infty \sqrt{x}e^{-x^2}dx \cdot \int_0^\infty \frac{e^{-x^2}}{\sqrt{x}}dx = \frac{\pi}{2\sqrt{2}}$ 	15	5	Medium	Easy	CO3	Descriptive

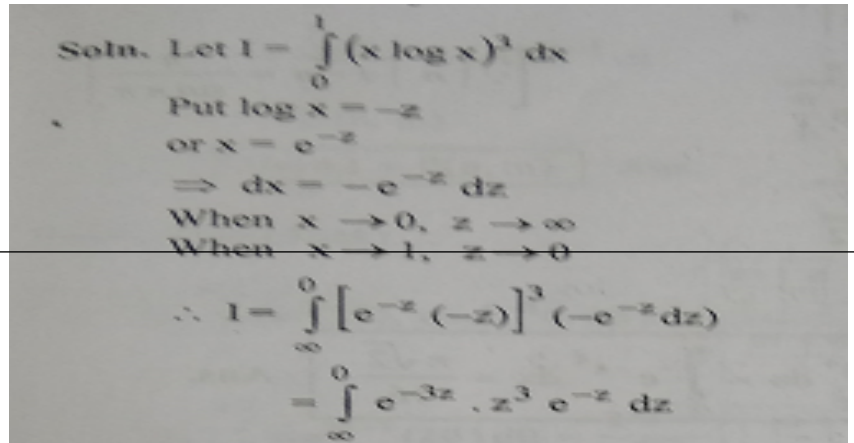
7	Prove that : $\int_0^\infty e^{-kx}\ x^{n-1}dx = \frac{\overline{|n}}{k^n}$ 	15	5	Medium	Easy	CO3	Descriptive

Soln. Let $I = \int_0^\infty e^{-kx}\ x^{n-1}\,dx$

Put $kx = t$

$\Rightarrow x = \frac{t}{k}$

$\therefore dx = \frac{1}{k}\,dt$

8	Evaluate : $\int_0^1 (x \log x)^3 dx$ 	15	5	Medium	Easy	CO3	Descriptive

Soln. Let $I = \int_0^1 (x \log x)^3 \, dx$

Put $\log x = -z$

or $x = e^{-z}$

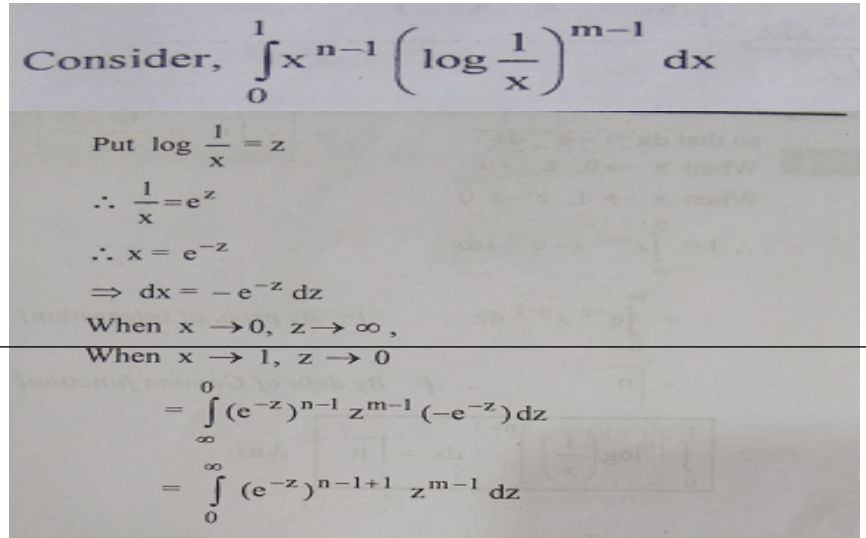
$\Rightarrow dx = -e^{-z} \, dz$

When $x \to 0$, $z \to \infty$

When $x \to 1$, $z \to 0$

$\therefore I = \int_\infty^0 \left[e^{-z}(-z)\right]^3 (-e^{-z} dz)$

$= \int_\infty^0 e^{-3z} \cdot z^3 \, e^{-z} \, dz$

9	Prove that $\int_0^1 x^{n-1}\left(\log\frac{1}{x}\right)^{m-1} dx = \frac{\lfloor m}{n^m}$ 	15	5	Medium	Easy	CO3	Descriptive

10	Prove that $\int_0^\infty \frac{x^a}{a^x}dx = \frac{\overline{\lvert a+1}}{(\log a)^{a+1}}$	15	5	Medium	Easy	CO3	Descriptive

Soln. Consider $\int_0^\infty \frac{x^a}{a^x}dx = \int_0^\infty \frac{x^a}{e^{\log a^x}}dx$ ..... (rearrangement)

$$= \int_0^\infty \frac{x^a}{e^{x\log a}}dx$$

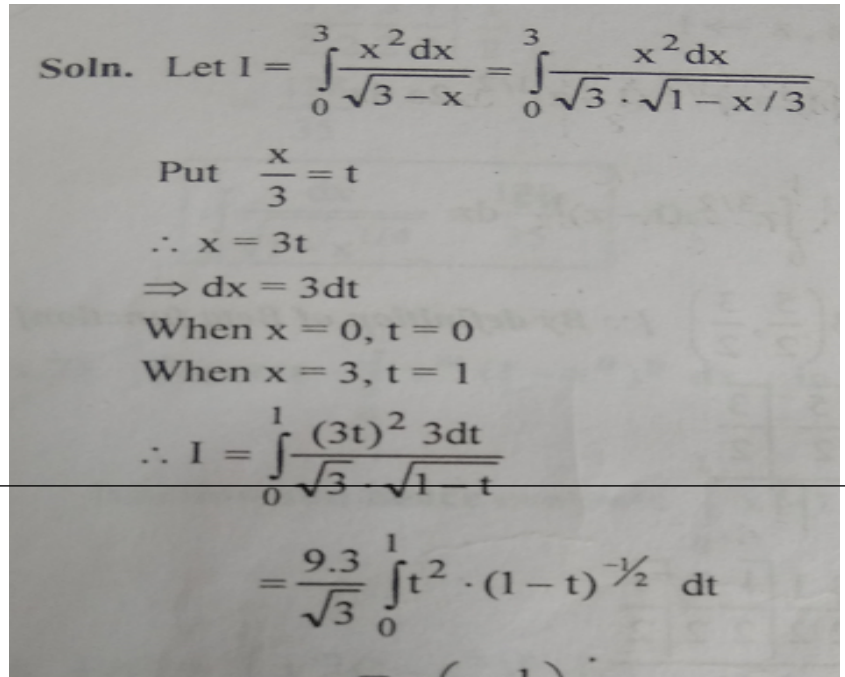
$$= \int_0^\infty x^a e^{-x\log a}dx$$

Put $x \log a = t$

$\therefore \quad x = \frac{t}{\log a}$

$\Rightarrow dx = \frac{dt}{\log a}$

When $x \to 0, \ t \to 0$

11	Evaluate $\int_0^3 \frac{x^2}{\sqrt{3-x}}dx$ 	15	5	Medium	Easy	CO3	Descriptive

Soln. Let $I = \int_0^3 \frac{x^2 dx}{\sqrt{3-x}} = \int_0^3 \frac{x^2 dx}{\sqrt{3}\cdot\sqrt{1-x/3}}$

Put $\frac{x}{3} = t$

$\therefore\ x = 3t$

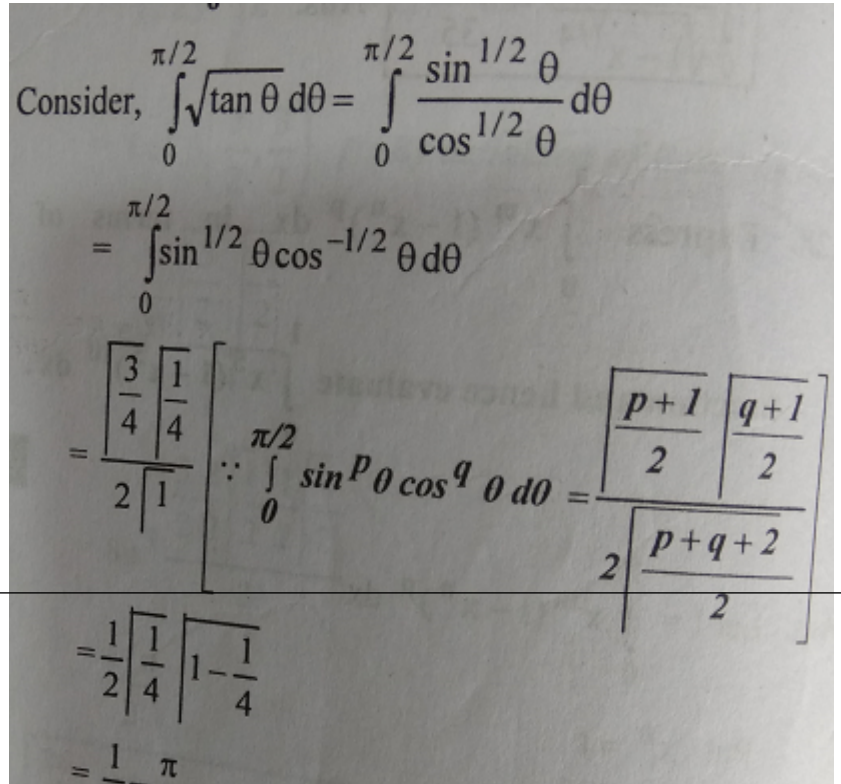
$\Rightarrow dx = 3dt$

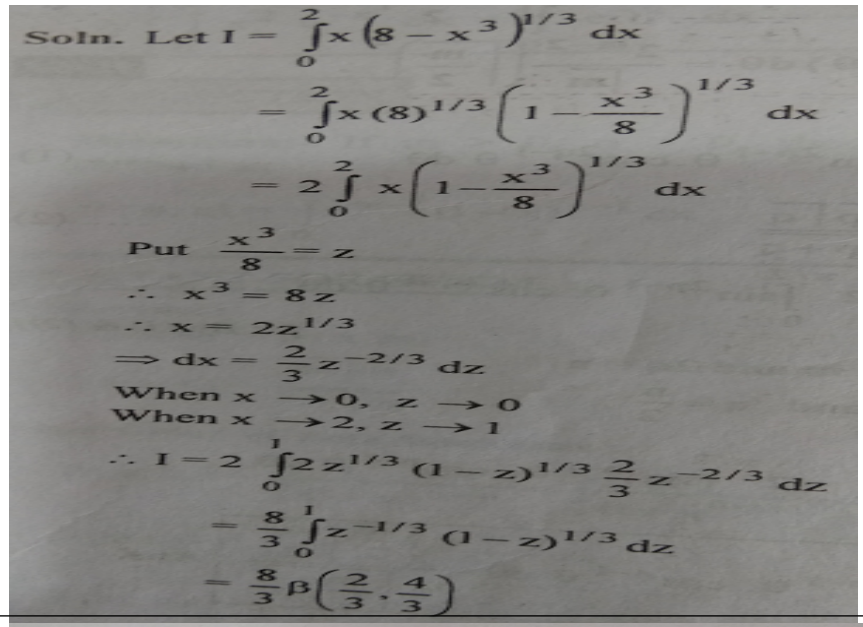
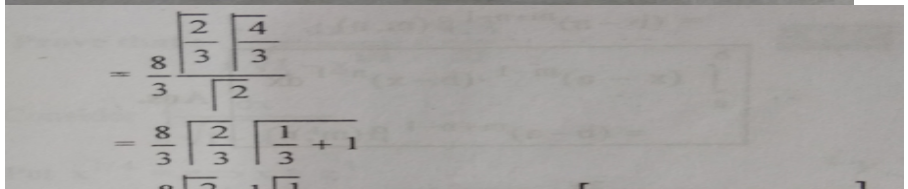
When $x = 0,\ t = 0$

When $x = 3,\ t = 1$

$\therefore\ I = \int_0^1 \frac{(3t)^2\ 3dt}{\sqrt{3}\cdot\sqrt{1-t}}$

$= \frac{9.3}{\sqrt{3}} \int_0^1 t^2 \cdot (1-t)^{-1/2}\ dt$

12	Show that $d\theta = \frac{\pi}{\sqrt{2}}$ 	15	5	Medium	Easy	CO3	Descriptive

13	Evaluate $\int_0^2 x\,(8-x^3)^{1/3}\,dx$  	15	5	Medium	Easy	CO3	Descriptive

Soln. Let $I = \int_0^2 x\,(8-x^3)^{1/3}\,dx$

$= \int_0^2 x\,(8)^{1/3}\left(1-\frac{x^3}{8}\right)^{1/3} dx$

$= 2\int_0^2 x\left(1-\frac{x^3}{8}\right)^{1/3} dx$

Put $\frac{x^3}{8} = z$

$\therefore\ x^3 = 8z$

$\therefore\ x = 2z^{1/3}$

$\Rightarrow dx = \frac{2}{3}z^{-2/3}\,dz$

When $x \to 0,\ z \to 0$

When $x \to 2,\ z \to 1$

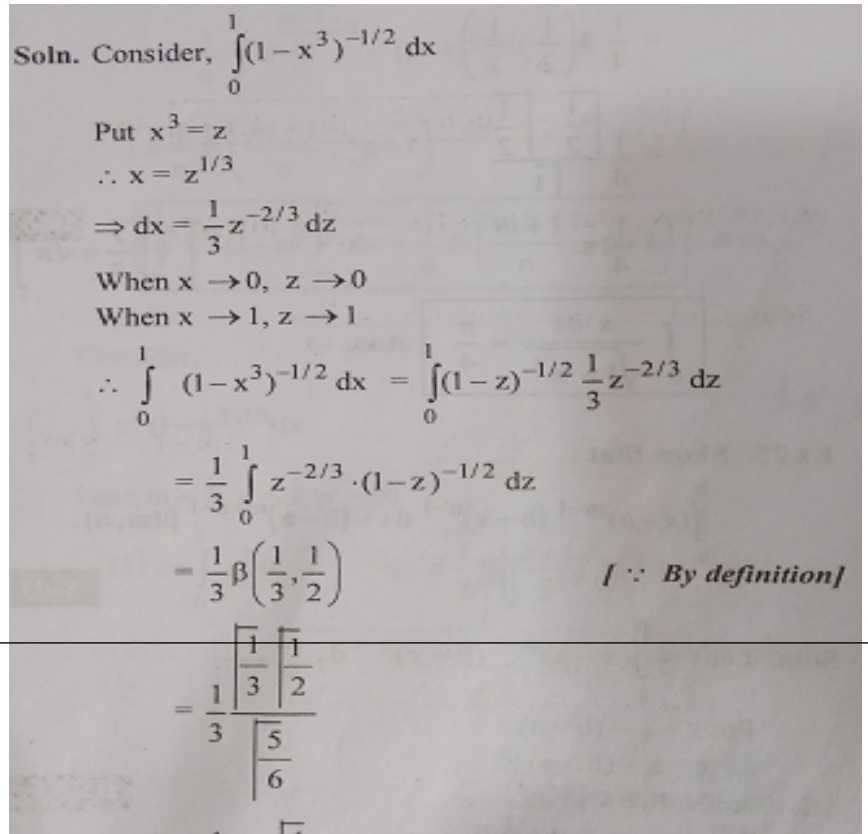
$\therefore\ I = 2\int_0^1 2z^{1/3}(1-z)^{1/3}\frac{2}{3}z^{-2/3}\,dz$

$= \frac{8}{3}\int_0^1 z^{-1/3}(1-z)^{1/3}\,dz$

$= \frac{8}{3}\beta\left(\frac{2}{3},\frac{4}{3}\right)$

$= \frac{8}{3}\frac{\lceil\frac{2}{3}\lceil\frac{4}{3}}{\lceil 2}$

$= \frac{8}{3}\lceil\frac{2}{3}\lceil\frac{1}{3}+1$

14	Evaluate $\int_0^1 (1-x^3)^{-1/2} dx$ 	15	5	Medium	Easy	CO3	Descriptive

Soln. Consider, $\int_0^1 (1-x^3)^{-1/2}\,dx$

Put $x^3 = z$

$\therefore x = z^{1/3}$

$\Rightarrow dx = \frac{1}{3} z^{-2/3}\,dz$

When $x \to 0,\ z \to 0$

When $x \to 1,\ z \to 1$

$$\therefore \int_0^1 (1-x^3)^{-1/2}\,dx = \int_0^1 (1-z)^{-1/2} \frac{1}{3} z^{-2/3}\,dz$$

$$= \frac{1}{3}\int_0^1 z^{-2/3}\cdot(1-z)^{-1/2}\,dz$$

$$= \frac{1}{3}\beta\left(\frac{1}{3},\frac{1}{2}\right) \qquad [\because \textit{By definition}]$$

$$= \frac{1}{3}\frac{\lceil\frac{1}{3}\lceil\frac{1}{2}}{\lceil\frac{5}{6}}$$

15	Evaluate $\int_0^1 \frac{x^2 dx}{\sqrt{1-x^4}} \cdot \int_0^1 \frac{dx}{\sqrt{1-x^4}}$.	15	5	Medium	Easy	CO3	Descriptive

Soln. Put $x^4 = z$

$\therefore x = z^{1/4}$

$dx = \frac{1}{4} z^{-3/4} dz$

When $x \to 0, z \to 0$ and when $x \to 1, z \to 1$

Now consider $\int_0^1 \frac{x^2 dx}{\sqrt{1-x^4}} \int_0^1 \frac{dx}{\sqrt{1-x^4}}$

$$= \int_0^1 \frac{(z^{1/4})^2 \frac{1}{4} z^{-3/4} dz}{(1-z)^{1/2}} \int_0^1 \frac{\frac{1}{4} z^{-3/4} dz}{(1-z)^{1/2}}$$

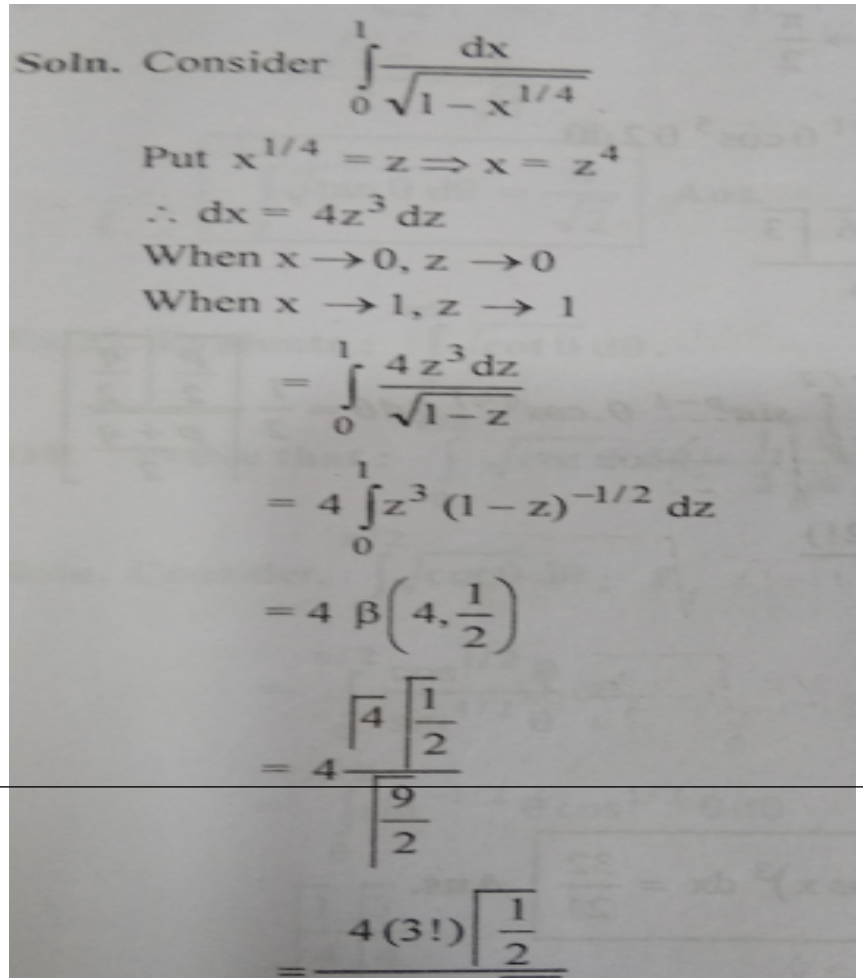
$$= \frac{1}{4} \int_0^1 \frac{z^{2/4} . z^{-3/4}}{(1-z)^{1/2}} dz . \frac{1}{4} \int_0^1 \frac{z^{-3/4}}{(1-z)^{1/2}} dz$$

$$= \frac{1}{16} \int_0^1 z^{-1/4} (1-z)^{-1/2} dz \int_0^1 z^{-3/4} (1-z)^{-1/2} dz$$

$$= \frac{1}{16} \beta\left(\frac{3}{4}, \frac{1}{2}\right) \cdot \beta\left(\frac{1}{4}, \frac{1}{2}\right) \quad [\because \text{By definition}]$$

$$= \frac{1}{16} \frac{\lceil\frac{3}{4} \lceil\frac{1}{2}}{\lceil\frac{5}{4}} \frac{\lceil\frac{1}{4} \lceil\frac{1}{2}}{\lceil\frac{3}{4}}$$

$$= \frac{\lceil\frac{1}{4}}{4} \pi$$

16	Evaluate $\int_0^1 \frac{1}{\sqrt{1-x^{1/4}}}\,dx$ 	15	5	Medium	Easy	CO3	Descriptive

Soln. Consider $\int_0^1 \frac{dx}{\sqrt{1-x^{1/4}}}$

Put $x^{1/4} = z \Rightarrow x = z^4$

$\therefore dx = 4z^3\,dz$

When $x \to 0,\ z \to 0$

When $x \to 1,\ z \to 1$

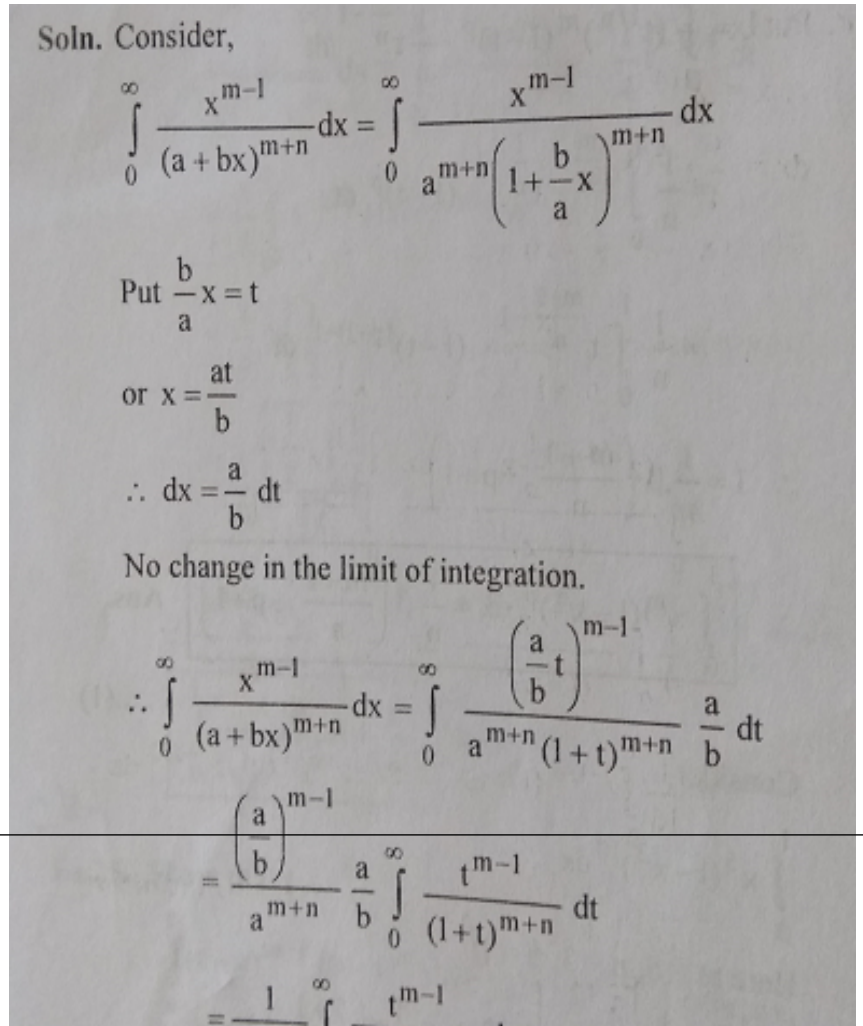
$$= \int_0^1 \frac{4z^3\,dz}{\sqrt{1-z}}$$

$$= 4\int_0^1 z^3 (1-z)^{-1/2}\,dz$$

$$= 4\,\beta\left(4, \frac{1}{2}\right)$$

$$= 4\frac{\lceil 4 \lceil \frac{1}{2}}{\lceil \frac{9}{2}}$$

$$= \frac{4(3!)\lceil \frac{1}{2}}{\lceil \frac{9}{2}}$$

17	Prove that $\int_0^{\infty} \frac{x^{m-1}}{(a+b\,x)^{m+n}}\,dx = \frac{\beta(m,n)}{a^n.b^m}$ 	15	5	Medium	Easy	CO3	Descriptive

Soln. Consider,

$$\int_0^{\infty} \frac{x^{m-1}}{(a+bx)^{m+n}}dx = \int_0^{\infty} \frac{x^{m-1}}{a^{m+n}\left(1+\frac{b}{a}x\right)^{m+n}}dx$$

Put $\frac{b}{a}x = t$

or $x = \frac{at}{b}$

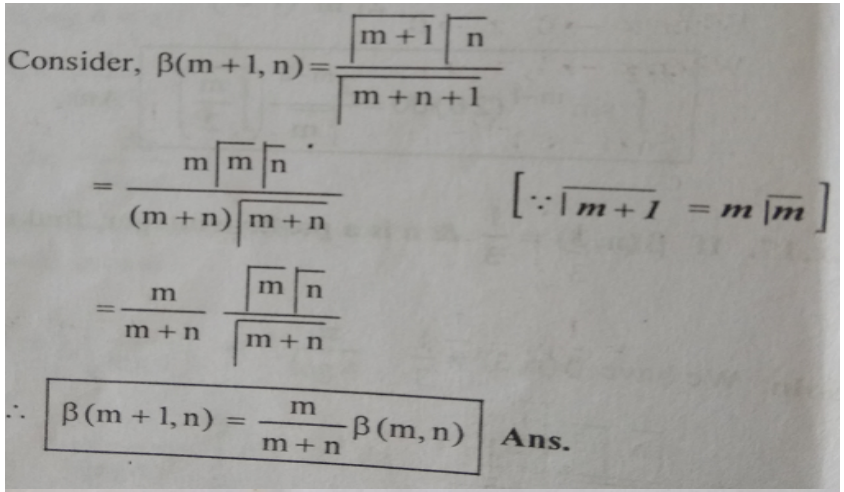
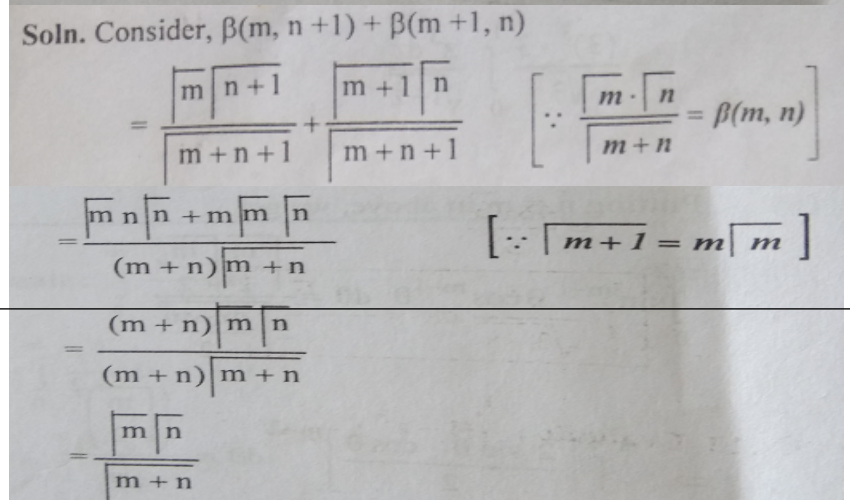
$\therefore\ dx = \frac{a}{b}\,dt$

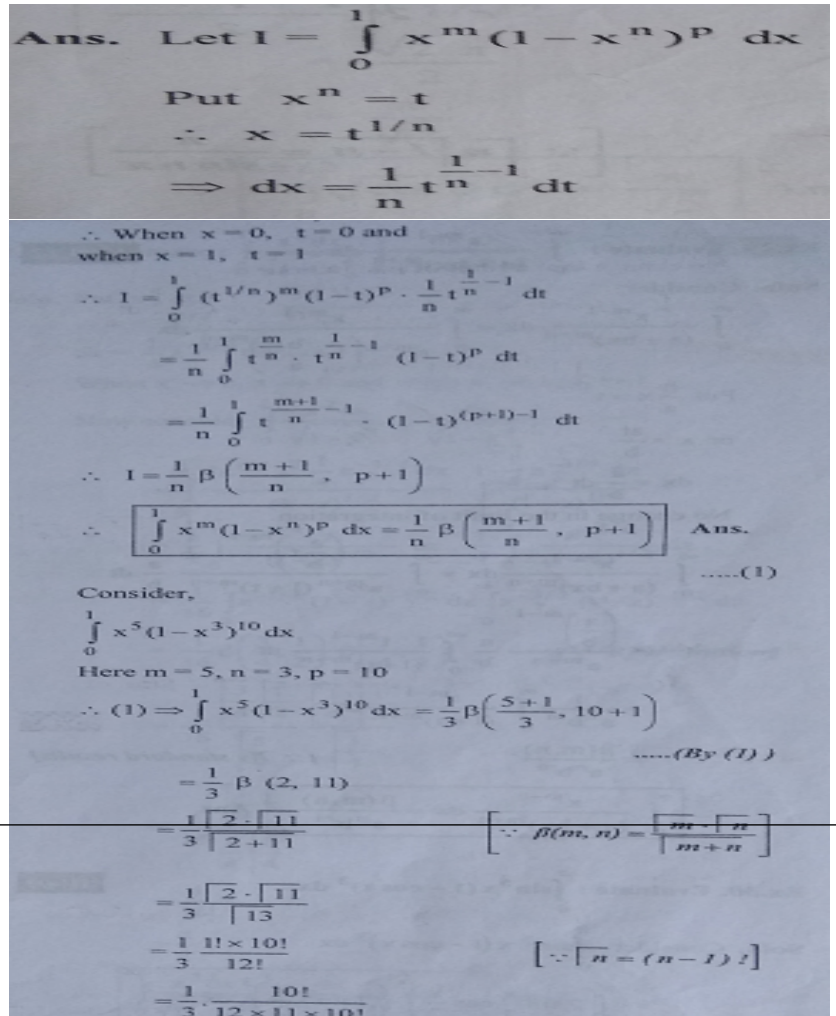
No change in the limit of integration.

$$\therefore \int_0^{\infty} \frac{x^{m-1}}{(a+bx)^{m+n}}dx = \int_0^{\infty} \frac{\left(\frac{a}{b}t\right)^{m-1}}{a^{m+n}(1+t)^{m+n}}\,\frac{a}{b}\,dt$$

$$= \frac{\left(\frac{a}{b}\right)^{m-1}}{a^{m+n}}\,\frac{a}{b}\int_0^{\infty} \frac{t^{m-1}}{(1+t)^{m+n}}\,dt$$

$$= \frac{1}{\int_0^{\infty}}$$

18	Prove that i) $\beta(m+1,n)=\frac{m}{m+n}\beta(m,n)$ $(ii)\ \beta(m,n)=\beta(m,n+1)+\beta(m+1,n)$  Consider, $\beta(m+1,n)=\frac{\lfloor m+1 \lfloor n}{\lfloor m+n+1}$ $=\frac{m\lfloor m \lfloor n}{(m+n)\lfloor m+n}$ $\left[\because \lfloor m+1 = m \lfloor m\right]$ $=\frac{m}{m+n}\frac{\lfloor m \lfloor n}{\lfloor m+n}$ $\therefore\ \boxed{\beta(m+1,n)=\frac{m}{m+n}\beta(m,n)}$ Ans.  Soln. Consider, $\beta(m, n+1)+\beta(m+1, n)$ $=\frac{\lfloor m \lfloor n+1}{\lfloor m+n+1}+\frac{\lfloor m+1 \lfloor n}{\lfloor m+n+1}$ $\left[\because \frac{\lfloor m \cdot \lfloor n}{\lfloor m+n}=\beta(m,n)\right]$ $=\frac{m\lfloor n \lfloor n+m\lfloor m \lfloor n}{(m+n)\lfloor m+n}$ $\left[\because \lfloor m+1 = m\lfloor m\right]$ $=\frac{(m+n)\lfloor m \lfloor n}{(m+n)\lfloor m+n}$ $=\frac{\lfloor m \lfloor n}{\lfloor m+n}$	15	5	Medium	Easy	CO3	Descriptive

19	Express $\int_0^1 x^m (1- x^n)^p dx$ in terms of beta function and hence evaluate $\int_0^1 x^5 (1-x^3)^{10} dx$ 	15	5	Medium	Easy	CO3	Descriptive

Ans. Let $I = \int_0^1 x^m (1-x^n)^p \, dx$

Put $x^n = t$

$\therefore \quad x = t^{1/n}$

$\Rightarrow dx = \frac{1}{n} t^{\frac{1}{n}-1} dt$

$\therefore$ When $x = 0, \quad t = 0$ and when $x = 1, \quad t = 1$

$$\therefore I = \int_0^1 (t^{1/n})^m (1-t)^p \cdot \frac{1}{n} t^{\frac{1}{n}-1} dt$$

$$= \frac{1}{n} \int_0^1 t^{\frac{m}{n}} \cdot t^{\frac{1}{n}-1} (1-t)^p \, dt$$

$$= \frac{1}{n} \int_0^1 t^{\frac{m+1}{n}-1} \cdot (1-t)^{(p+1)-1} \, dt$$

$$\therefore \quad I = \frac{1}{n} \beta\left(\frac{m+1}{n}, \; p+1\right)$$

$$\therefore \quad \boxed{\int_0^1 x^m (1-x^n)^p \, dx = \frac{1}{n} \beta\left(\frac{m+1}{n}, \; p+1\right)} \quad \text{Ans.}$$

.....(1)

Consider,

$$\int_0^1 x^5 (1-x^3)^{10} dx$$

Here $m = 5, n = 3, p = 10$

$$\therefore (1) \Rightarrow \int_0^1 x^5 (1-x^3)^{10} dx = \frac{1}{3} \beta\left(\frac{5+1}{3}, 10+1\right)$$

.....(By (1))

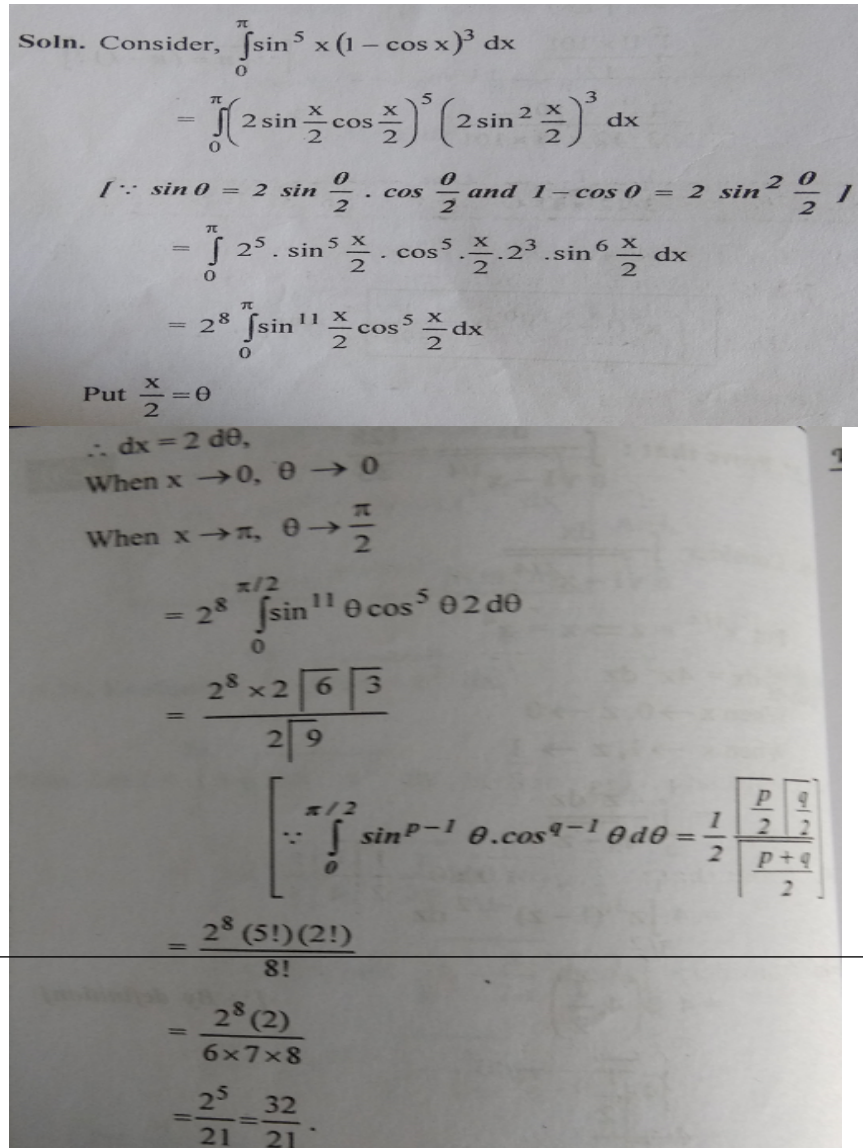
$$= \frac{1}{3} \beta \, (2, \; 11)$$

$$= \frac{1}{3} \frac{\lceil 2 \cdot \lceil 11}{\lceil 2+11} \qquad \left[\because \beta(m, n) = \frac{\lceil m \cdot \lceil n}{\lceil m+n}\right]$$

$$= \frac{1}{3} \frac{\lceil 2 \cdot \lceil 11}{\lceil 13}$$

$$= \frac{1}{3} \frac{1! \times 10!}{12!} \qquad [\because \lceil n = (n-1)!]$$

$$= \frac{1}{3} \cdot \frac{10!}{12 \times 11 \times 10!}$$

20	Evaluate $\int_0^{\pi} \sin^5 x(1-\cos x)^3 dx$ 	15	5	Medium	Easy	CO3	Descriptive

Soln. Consider, $\int_0^{\pi} \sin^5 x(1-\cos x)^3 dx$

$$= \int_0^{\pi} \left(2\sin\frac{x}{2}\cos\frac{x}{2}\right)^5 \left(2\sin^2\frac{x}{2}\right)^3 dx$$

$$= \int_0^{\pi} 2^5 . \sin^5\frac{x}{2} . \cos^5\frac{x}{2} . 2^3 . \sin^6\frac{x}{2} dx$$

$$= 2^8 \int_0^{\pi} \sin^{11}\frac{x}{2}\cos^5\frac{x}{2} dx$$

Put $\frac{x}{2} = \theta$

$\therefore dx = 2\,d\theta$,

When $x \to 0$, $\theta \to 0$

When $x \to \pi$, $\theta \to \frac{\pi}{2}$

$$= 2^8 \int_0^{\pi/2} \sin^{11}\theta\cos^5\theta\, 2\,d\theta$$

$$= \frac{2^8 \times 2\lfloor 6 \lfloor 3}{2\lfloor 9}$$

$$\left[\because \int_0^{\pi/2} \sin^{p-1}\theta.\cos^{q-1}\theta\, d\theta = \frac{1}{2}\frac{\lfloor\frac{p}{2}\lfloor\frac{q}{2}}{\lfloor\frac{p+q}{2}}\right]$$

$$= \frac{2^8(5!)(2!)}{8!}$$

$$= \frac{2^8(2)}{6\times7\times8}$$

$$= \frac{2^5}{21} = \frac{32}{21}.$$

21	Evaluate $\int_0^\infty \sqrt{x}e^{-\sqrt[3]{x}}dx$	15	5	Medium	Easy	CO3	Descriptive

	Q. Evaluate $\int_0^\infty \sqrt{x}\, e^{-3\sqrt{x}}\, dx$ Sol^n Put $\sqrt[3]{x} = t$ $\Rightarrow x^{1/3} = t \Rightarrow x = t^3$ Diff^n w.r.t to x, $\frac{1}{3} x^{-\frac{2}{3}} dx = dt$ $\Rightarrow dx = 3x^{2/3} dt$ $\Rightarrow dx = 3t^2 dt$ $\therefore \int_0^\infty \sqrt{x}\, e^{-3\sqrt{x}} dx = \int_0^\infty t^{3/2} e^{-t} 3t^2 dt$ $= 3\int_0^\infty e^{-t} t^{\frac{7}{2}} dt$ $= 3\int_0^\infty e^{-t} t^{\frac{9}{2}-1} dt$ $= 3\,\Gamma(9/2)$ $= 3 \cdot \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}$ $= \frac{315\sqrt{\pi}}{16}$						
22	Prove that $\int_0^\infty \frac{x^7}{7^x} dx = \frac{7!}{(\log 7)^8}$	15	5	Medium	Easy	CO3	Descriptive

Sol^{n}

$$7^x = e^{\log 7^x} = e^{x \log 7}$$

$$\therefore \int_0^\infty \frac{x^7}{7^x}\, dx = \int_0^\infty e^{-x \log 7}\, x^7\, dx$$

Put $x \log 7 = t$

$$\Rightarrow dx = \frac{dt}{\log 7}$$

$$\therefore \int_0^\infty \frac{x^7\, dx}{7^x} = \int_0^\infty e^{-t} \left(\frac{t}{\log 7}\right)^7 \frac{dt}{\log 7}$$

$$= \frac{1}{(\log 7)^8} \int_0^\infty e^{-t}\, t^7\, dt$$

$$= \frac{1}{(\log 7)^8} \boxed{8}$$

$$= \frac{7!}{(\log 7)^8} \qquad \left\{ \because \boxed{n+1} = \right.$$